

INTERVIEW PREP · FINANCE ROLES

Financial Modeling & Excel

The three-statement model, supporting schedules, circularity, scenarios, Excel mastery, and LBO/DCF/M&A model types.

Subject 04 of the Finance Interview Series · Target roles: Equity Research · Investment Banking · FP&A · Product Control

Audience: MBA (Finance) fresher — B.E. Computer Science, ~2 years on a derivatives desk, NISM Series-XV certified

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1. Why Modeling Matters & How Interviews Test It

A financial model is a spreadsheet that projects a company's financial performance into the future so that a decision — a buy/sell rating, a valuation, a financing package, a budget, or an acquisition price — can be made on numbers rather than intuition. In equity research it feeds a target price; in investment banking it supports a pitch, an LBO, or a merger; in FP&A it becomes the annual operating plan and rolling forecast; in product control it validates the P&L that a trading desk booked. Across all four of your target roles the same core artefact appears again and again: the **integrated three-statement model**.

Interviewers rarely ask you to build a full model on the spot (that is what modeling tests and case studies are for). Instead they probe whether you *understand the plumbing*: how a change in one number ripples through all three statements, why the balance sheet balances, how you would find a mistake, and whether your Excel is disciplined. If you can walk fluently through the mechanics — depreciation flowing from the fixed-asset schedule into the P&L, into accumulated depreciation on the balance sheet, and back out as a non-cash add-back in the cash flow — you signal that you have actually built models, not just watched a video.

This guide is written to make you fluent in exactly that plumbing, and to give you crisp, correct Excel syntax you can quote under pressure. Everything is expressed in a way that transfers directly to Indian companies and INR, because your interviews will likely reference Nifty/BSE names, Ind AS financials, and rupee figures in crore (₹ crore = ₹10 million) and lakh (₹1 lakh = ₹100,000).

Key takeaways: A model exists to support a decision. The three-statement model is the backbone of DCF, LBO, and M&A models alike. Interviews test your understanding of the *links* and your Excel discipline far more than your typing speed.

2. The Three-Statement Model — Architecture

The three-statement model integrates the **Income Statement (P&L)**, the **Balance Sheet**, and the **Cash Flow Statement** into a single, dynamically linked workbook. "Integrated" means that when you change one assumption — say revenue growth from 12% to 15% — every dependent line on all three statements updates automatically, and the balance sheet still balances. That property is the entire point; a model where the statements are typed independently is not a model, it is three disconnected tables.

2.1 The standard layout

A clean model separates **inputs**, **calculations**, and **outputs**. A typical tab structure:

- **Cover / Contents** — model name, author, version, currency (INR crore), units, and a change log.
- **Assumptions / Drivers** — every hardcoded input in one place, colour-coded blue.
- **Income Statement** — revenue down to net income.
- **Balance Sheet** — assets, liabilities, equity.
- **Cash Flow Statement** — operating, investing, financing.
- **Supporting schedules** — revenue build, depreciation (PP&E), debt & interest, working capital, equity roll-forward. Often each on its own tab or in a clearly boxed section.
- **Outputs** — ratios, DCF/valuation, scenario summary, charts.

Columns run left-to-right in time: historical actuals (usually 3 years) followed by forecast years (usually 5). A common convention is to shade historical columns differently and to place a single-cell "years of history / years of forecast" control near the top.

2.2 The build order

You do not build the three statements top-to-bottom in isolation. The professional order is:

1. **Input the historicals** for all three statements from the annual report / Ind AS financials. Make them tie out first.
2. **Build the supporting schedules** off the assumptions: revenue build, cost build, depreciation, debt, working capital.
3. **Project the Income Statement** down to EBIT, then pause — interest depends on the debt schedule.
4. **Project the Balance Sheet** line items that are driven by schedules (PP&E from the depreciation schedule, working capital from the WC schedule, debt from the debt schedule, equity from the roll-forward).
5. **Build the Cash Flow Statement**, which pulls from the P&L (net income, D&A) and from the change in balance sheet items.
6. **Link the ending cash** from the cash flow into the balance sheet cash line; link the revolver/interest back into the P&L. This closes the loop and typically creates the intended circularity.
7. **Check the balance**: Total Assets – (Total Liabilities + Equity) must equal zero in every column.

Common trap: Building the cash flow statement by typing numbers instead of *deriving* it from the P&L and the change in balance-sheet accounts. If your cash flow is not mechanically linked to the other two statements, the model will never self-check and you have lost the single most valuable diagnostic in the whole workbook.

3. Linking P&L → Balance Sheet → Cash Flow

The three statements are welded together at specific, memorisable joints. Master these and you can answer almost any "how are the statements linked" question.

3.1 The core links

Link	From	To	Mechanics
Net income	P&L (bottom line)	Cash Flow (top of CFO) & Retained Earnings (BS equity)	Net income starts the indirect cash flow and rolls into retained earnings.
Depreciation & Amortisation	Depreciation schedule → P&L expense	Cash Flow (add-back) & Accumulated Depreciation (BS)	Non-cash: reduces P&L profit, added back in CFO, reduces net PP&E on BS.
Capex	Assumptions / capex schedule	Cash Flow (CFI outflow) & Gross PP&E (BS)	Cash out in investing; increases gross fixed assets.
Change in working capital	Working-capital schedule (Δ receivables, inventory, payables)	Cash Flow (CFO)	Increase in an asset uses cash; increase in a liability is a source of cash.
Debt drawdowns / repayments	Debt schedule	Cash Flow (CFF) & Debt (BS)	Draw = cash in + debt up; repay = cash out + debt down.
Interest expense	Debt schedule (rate × balance)	P&L (below EBIT)	Reduces pre-tax income; source of the classic circularity.
Interest income	Cash / investments × rate	P&L	Also feeds circularity via the cash balance.
Dividends / buybacks	Assumptions / equity schedule	Cash Flow (CFF) & Retained Earnings / Share capital (BS)	Cash out; reduces equity.
Ending cash	Cash Flow (last line)	Balance Sheet (cash line)	The single most important link — the cash flow's ending cash is the balance sheet's cash.

3.2 The mental model: the cash flow statement is the bridge

The cleanest way to remember the linkages is that the cash flow statement **reconciles the beginning and ending balance sheet**. Net income comes from the P&L. Every other line of the cash flow statement is either a non-cash item from the P&L that must be reversed (D&A, stock-based comp), or the *change* in a balance-sheet account (receivables, inventory, payables, debt, equity). Add net income to all those adjustments and you get the change in cash — which, added to opening cash, gives closing cash, which flows straight into the balance sheet.

Indirect-method Cash Flow (skeleton)

$$\text{CFO} = \text{Net income} + \text{D\&A} + \text{other non-cash} - \Delta(\text{operating assets}) + \Delta(\text{operating liabilities})$$

$$\text{CFI} = -\text{Capex} + \text{asset sales} - \text{acquisitions} + \Delta(\text{investments})$$

$$\text{CFF} = +\text{Debt drawn} - \text{Debt repaid} + \text{Equity issued} - \text{Buybacks} - \text{Dividends}$$

$$\Delta \text{Cash} = \text{CFO} + \text{CFI} + \text{CFF}$$

$$\text{Ending cash} = \text{Beginning cash} + \Delta \text{Cash}$$

Sign rule: an increase in an *asset* uses cash (negative); an increase in a *liability or equity* is a source of cash (positive).

Worked Example 1 — One assumption ripples through all three statements

Suppose you increase forecast depreciation by ₹100 (crore), tax rate 25%. Trace it:

- **P&L:** EBIT falls by ₹100. Pre-tax income falls ₹100. Tax falls by ₹25 ($25\% \times 100$). Net income falls by ₹75.
- **Cash Flow (CFO):** Net income $-\text{₹}75$, but we add back the full ₹100 of D&A. Net effect on CFO = + ₹25 (the tax shield). Cash rises ₹25.
- **Balance Sheet:** Accumulated depreciation up ₹100 $\rightarrow$ net PP&E down ₹100. Retained earnings down ₹75. Cash up ₹25.
- **Balance check:** Assets change = -100 (PP&E) + 25 (cash) = -75 . Equity change = -75 . Liabilities unchanged. Both sides move by $-75 \rightarrow$ still balances. ✓

This "trace it through" drill is the single most common technical modeling question. Practise it for depreciation, an inventory purchase, a debt draw, and a dividend until it is automatic.

4. The Cash & Revolver Plug — How the Model Balances

Once revenue, costs, capex, working capital, and scheduled debt are all projected, the model produces a forecast cash balance. But two problems arise: (1) what if the company generates more cash than it needs? (2) what if it runs short and cannot fund its obligations? The **cash sweep and revolver** mechanism handles both and is the device that makes the model self-balance in the financing section.

4.1 The revolving credit facility (revolver) as the plug

A revolver is a flexible line of credit. In the model it acts as the **plug**: if the company has a cash shortfall (cash would fall below a required minimum), it *draws* on the revolver; if it has surplus cash, it *repays* the revolver (a "cash sweep") before letting cash pile up. The logic:

Revolver logic

Cash available before revolver = Beginning cash + CFO + CFI + CFF(excl. revolver)

If (Cash available < Minimum cash) → **Draw** = Minimum cash – Cash available

If (Cash available > Minimum cash) → **Repay** = min(Cash surplus, Revolver balance outstanding)

Ending cash = max(Cash available, Minimum cash) after the draw/repay

Minimum cash is a policy assumption (e.g. ₹50 crore operating buffer). The revolver can never be repaid below zero.

Worked Example 2 — Excel revolver formulas

Assume: B10 = cash available before revolver, B11 = minimum cash, B12 = beginning revolver balance.

Revolver draw (only when short of the minimum):

=MAX(0, B11 - B10)

Revolver repayment (sweep surplus, but never more than what is outstanding):

=MIN(B12, MAX(0, B10 - B11))

Ending revolver balance:

=B12 + Draw - Repayment

Ending cash:

=B10 + Draw - Repayment

Interest on the revolver then feeds back into the P&L — that is the classic circular reference discussed in Section 7.

Common trap: Plugging cash the wrong way — hardcoding a number into the balance-sheet cash line to force it to balance. That masks a real error. Cash must always come from the cash flow statement's ending balance. The revolver, not a hardcoded, is the legitimate financing plug.

4.2 What "plug" means and does not mean

"Plug" is banker slang for the item that absorbs whatever is left over so the model closes. The *legitimate* plugs are cash (asset side) and the revolver/short-term debt (liability side). An *illegitimate* plug is any balancing figure you jam into an unrelated line just to make the two sides equal — that is hiding a mistake, not modeling.

5. Why the Balance Sheet Must Balance (and Debugging It)

5.1 Why it balances at all

The balance sheet balances because of the fundamental accounting identity and double-entry bookkeeping:

The accounting identity

$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$

Every transaction affects at least two accounts such that the identity is preserved. Equity is the residual claim on assets after liabilities.

In a correctly linked model the identity is preserved *automatically* because of the way the cash flow statement reconciles the two statements. Each cash-flow line is either a P&L non-cash reversal or the change in a balance-sheet account. When ending cash (which incorporates all those changes) flows into the balance sheet, and retained earnings absorbs net income minus dividends, the two sides move in lockstep. If your model does *not* balance, you have a linking error — not a fundamental accounting impossibility.

5.2 A disciplined balance check

Every model should have a permanent **balance check row**: $\text{Total Assets} - \text{Total Liabilities} - \text{Total Equity}$, formatted to flash red if it is not zero (use conditional formatting on $\text{ABS}(\text{check}) > 0.5$). Build it once and watch it as you extend the model.

5.3 Debugging an out-of-balance model — systematic method

When the balance check is non-zero, do not randomly poke cells. Work methodically:

1. **Find the first year that breaks.** If years 1–2 balance and year 3 breaks, the error was introduced in year 3's formulas.
2. **Note the size and sign of the imbalance.** The exact amount is a huge clue — search the model for that number (Ctrl+F). If assets exceed L+E by ₹75 and depreciation this year is ₹75, you have a depreciation link problem.
3. **Check the cash flow statement first.** The most common cause is a balance-sheet line that changed but whose change was not captured in the cash flow (e.g. you added "other assets" on the BS but forgot the corresponding CFO line). Ensure every non-cash BS line has a corresponding cash-flow line.
4. **Verify sign conventions.** An increase in receivables should *reduce* cash. A flipped sign typically produces an imbalance of exactly twice the change.
5. **Confirm ending cash links.** The BS cash line must equal the cash flow statement's ending cash, not a stale reference.
6. **Check retained earnings roll-forward.** $\text{RE}(t) = \text{RE}(t-1) + \text{Net income} - \text{Dividends}$. A common bug is linking to the wrong year or omitting dividends.
7. **Check that each schedule ties back.** PP&E on the BS = ending net PP&E from the depreciation schedule; debt on the BS = ending debt from the debt schedule.

Common trap: An imbalance of exactly 2× some line item almost always means a **sign error** (you added where you should have subtracted). An imbalance equal to a single line item usually means a **missing link** (a BS change with no matching cash-flow entry).

Key takeaways: The balance sheet balances by construction when links are correct. Keep a permanent balance-check row. Debug by finding the first broken year, matching the imbalance amount to a line item, and confirming that every balance-sheet change has a matching cash-flow entry with the right sign.

6. Supporting Schedules

Supporting schedules are the engine room. They convert clean assumptions into the detailed line items the three statements consume. Building schedules separately keeps the statements themselves simple (each cell just references a schedule) and makes the logic auditable.

6.1 Revenue build (the driver tree)

Never forecast revenue as a single "grows 10%" line if you can decompose it. A driver-based build ties revenue to **volume × price**, or **units × ASP**, or **customers × ARPU**, or for a bank, **average assets × yield**. This lets you flex the real levers in scenarios.

There are two directions to a revenue build, and a good analyst reconciles them:

- **Top-down:** start from the total addressable market (TAM), apply a market-share assumption, and work down to the company's revenue. Example: Indian two-wheeler market ≈ 18 million units $\times$ company market share 15% $\times$ ASP ₹75,000. Good for framing a growth story and for early-stage or new-market forecasts, but share assumptions can be hand-wavy.
- **Bottom-up:** build from the company's own capacity and operations — stores $\times$ sales-per-store, plants $\times$ utilisation $\times$ output, salesforce $\times$ deals $\times$ ticket size, or existing customers + net adds. More grounded and defensible; the preferred method when you have segment disclosure.

Best practice: build bottom-up, then cross-check the implied market share top-down for a sanity test. If your bottom-up forecast implies the company captures 60% of a market it holds 15% of today, the forecast is too aggressive.

Revenue driver examples

Manufacturer: Revenue = Units sold $\times$ Average selling price

SaaS: Revenue = Customers $\times$ ARPU (or Beginning ARR + New – Churn)

Retail/QSR: Revenue = Number of stores $\times$ Sales per store

Bank NII: Net interest income = Average interest-earning assets $\times$ Net interest margin

Worked Example 3 — A small revenue driver build

A two-wheeler maker. Assumptions in column C (year 1): units C5 = 1,000,000, ASP C6 = ₹75,000. Volume grows 8% p.a., ASP grows 4% p.a.

- Year-1 revenue (in ₹): $=C5 \times C6 \rightarrow ₹75,000$ crore-equivalent (scale as needed). Keep units consistent.
- Year-2 units: $=C5 \times (1+B8)$ where B8 = 8% volume growth (absolute reference so it copies across).
- Year-2 ASP: $=C6 \times (1+B9)$ where B9 = 4% price growth.
- Year-2 revenue: $=D5 \times D6$ — one formula pattern copied across every forecast column.

Now a scenario can flex volume growth and ASP independently — far more insightful than a blanket revenue growth rate.

6.2 Cost build

Split costs into **variable** (scale with revenue or volume — e.g. raw material as % of sales) and **fixed** (grow with inflation or step up with capacity). Model COGS often as a % of revenue (gross-margin driven); model employee cost as headcount $\times$ average cost; model other opex with an inflation escalator. Keep each cost line referencing an explicit driver on the assumptions tab.

Cost build patterns

COGS = Revenue $\times$ (1 – Gross margin %) or Units $\times$ cost per unit

Employee cost = Headcount $\times$ average cost per employee $\times$ (1 + increment %)

Other opex = Prior-year opex $\times$ (1 + inflation %) [semi-fixed]

6.3 Depreciation schedule (existing + new capex)

Depreciation must be split between **existing assets** (depreciate the opening net book value on a declining or remaining-life basis) and **new capex** (each year's additions depreciate over their useful life). The schedule outputs (a) total depreciation $\rightarrow$ P&L and CFO add-back, and (b) ending net PP&E $\rightarrow$ balance sheet.

PP&E roll-forward

Ending gross PP&E = Beginning gross PP&E + Capex – Disposals (at cost)

Ending accumulated dep. = Beginning acc. dep. + Depreciation – Disposal dep.

Ending net PP&E = Ending gross – Ending accumulated depreciation

Simplified (net basis): Ending net PP&E = Beginning net PP&E + Capex – Depreciation

The simplified net-basis roll-forward is the one most commonly used in a first pass and in interviews.

Worked Example 4 — A straight-line depreciation schedule

Existing net PP&E ₹1,000 crore, remaining life 10 years $\rightarrow$ existing depreciation ₹100/yr. New capex ₹200/yr, useful life 5 years $\rightarrow$ ₹40/yr per vintage, applied from the year of spend.

Item (₹ cr)	Yr 1	Yr 2	Yr 3
Dep. on existing	100	100	100
Dep. on Yr-1 capex	40	40	40
Dep. on Yr-2 capex	—	40	40
Dep. on Yr-3 capex	—	—	40
Total depreciation	140	180	220
Beginning net PP&E	1,000	1,060	1,080
+ Capex	200	200	200
– Depreciation	140	180	220
Ending net PP&E	1,060	1,080	1,060

A quick shortcut when you cannot build a full vintage schedule: model depreciation as a % of opening gross block or as a % of revenue, calibrated to the historical ratio. Interviewers accept this if you state the simplification.

6.4 Debt & interest schedule

For each tranche (term loan, bonds, revolver), roll the balance forward: beginning + drawdowns – scheduled repayments (amortisation) – optional prepayments = ending. Interest = rate $\times$ balance. Best practice is to charge interest on the **average** of beginning and ending balance, which is more accurate but creates circularity (Section 7).

Debt & interest

Ending debt = Beginning debt + Draws – Mandatory repayments – Optional prepayments

Interest expense = Interest rate $\times$ Average debt = Rate $\times$ (Beginning + Ending) / 2

(Non-circular alternative) Interest = Rate $\times$ Beginning balance

Charging on beginning balance avoids circularity at a small cost in precision — a common interview-safe choice.

6.5 Working-capital schedule

Working capital is driven off activity ratios (days). Forecast each item using its ratio, then the *change* in each flows to the cash flow statement.

Working-capital drivers (days method)

Receivables = (DSO / 365) $\times$ Revenue [DSO = days sales outstanding]

Inventory = (DIO / 365) $\times$ COGS [DIO = days inventory outstanding]

Payables = (DPO / 365) $\times$ COGS [DPO = days payables outstanding]

Cash conversion cycle = DSO + DIO – DPO

Δ Net working capital $\rightarrow$ cash flow (an increase in NWC uses cash)

6.6 Equity / retained-earnings roll-forward**Equity roll-forward**

Ending retained earnings = Beginning RE + Net income – Dividends – Buybacks (RE portion)

Ending share capital = Beginning + New issuance – Buybacks (par portion)

Total equity = Share capital + Reserves & surplus (incl. RE) + Other comprehensive income

Under Ind AS, "Reserves & Surplus" houses retained earnings; dividends reduce it in the period declared.

Key takeaways: Schedules turn assumptions into statement line items. Build revenue and cost off real drivers; split depreciation into existing vs new capex; roll debt and equity forward; drive working capital off days ratios. Each statement line should reference a schedule, not a hardcode.

7. Circular References & Iterative Calculation

7.1 Where circularity comes from

The classic circular reference in a three-statement model runs through **interest**:

Interest expense depends on the debt balance and cash balance → interest affects net income → net income affects cash flow → cash flow affects the cash and revolver balances → which affect interest. The loop closes on itself. The same loop appears when interest is charged on *average* debt (ending balance depends on the cash generated, which depends on interest, which depends on the ending balance).

The interest circularity

Interest → Net income → Cash flow → Cash / Revolver balance → Average debt → Interest → ...

A legitimate, intentional circularity. Excel resolves it by iterating until the numbers converge.

7.2 Iterative calculation

Excel normally forbids circular references and shows a warning. To let the interest loop resolve, enable iterative calculation: **File** → **Options** → **Formulas** → **Enable iterative calculation** (set Maximum Iterations ~100, Maximum Change 0.001). Excel then recomputes the loop repeatedly until values stop changing — they *converge* to the consistent answer.

7.3 The problem with leaving iteration on

Iterative calc is fragile. If any cell in the loop ever returns an error (e.g. a #DIV/0! during editing), the error propagates around the circle and can freeze the whole model at #REF! or 0, and turning iteration off then on may not clear it. Models can also become slow. Professionals therefore build a **circularity switch**.

7.4 The circularity switch (circuit breaker)

Add a single input cell, e.g. Circ_Switch = 1 (on) or 0 (off). Wrap the interest calculation so that when the switch is 0, interest is forced to zero (or to a hardcoded prior value), breaking the loop and clearing any propagated error. Flip it back to 1 to restore live calculation.

Worked Example 5 — Circularity switch & average-debt interest

Let \$B\$2 = circularity switch (1/0), interest rate in \$B\$3, beginning debt in C10, ending debt in C11.

Interest on average debt, protected by the switch:

=IF(\$B\$2=1, \$B\$3*(C10+C11)/2, 0)

When the model breaks (an error races round the loop), set B2 = 0: interest becomes 0, the loop opens, errors clear, and the model recalculates. Set B2 = 1 to resume the live average-debt interest. Wrapping with IFERROR as a second layer is also common:

=IFERROR(IF(\$B\$2=1, \$B\$3*(C10+C11)/2, 0), 0)

7.5 Alternatives to iteration

- **Charge interest on the opening balance** only — removes the circularity entirely at the cost of minor precision. Many buy-side and interview models do exactly this.
- **Copy-paste values** — compute interest once with iteration on, then paste the resulting interest figures as hardcoded values and turn iteration off. Fast and stable, but the interest no longer updates when

assumptions change (you must repeat the paste). Acceptable for a one-off; risky in a live model.

- **Macro / VBA copy-paste loop** — a button that runs the iteration in code and pastes values. Robust but adds complexity.

Common trap: Forgetting to switch iterative calculation back on after opening a colleague's model, then panicking at a circular-reference warning. Also: leaving a "copy-paste values" interest hardcode in place and wondering why interest never changes when you flex leverage. Always know whether your interest is live or pasted.

8. Scenario & Sensitivity Analysis

A model's value multiplies when it can answer "what if?". Three tools: **toggles/scenarios**, **data tables**, and the **Scenario Manager**.

8.1 Scenario toggles with CHOOSE / INDEX

Build Base / Bull / Bear assumption sets in three rows, then a single **scenario selector cell** (1, 2, or 3) that picks the live set. CHOOSE or INDEX reads the selector and returns the matching assumption; every driver references the selected row so the whole model flips with one cell.

Worked Example 6 — Scenario toggle

Selector in \$B\$1 (1 = Base, 2 = Bull, 3 = Bear). Revenue growth for the three cases sits in C5 (Base), C6 (Bull), C7 (Bear).

Live revenue growth:

=CHOOSE(\$B\$1, C5, C6, C7)

Equivalent with INDEX (scales better to many drivers — put the selector as the row offset):

=INDEX(C5:C7, \$B\$1)

Combine with a Form Control drop-down (Developer → Insert → Combo/Spin) linked to B1 so a user just picks "Bull" from a menu.

8.2 Data tables — one-variable and two-variable

A **Data Table** (Data → What-If Analysis → Data Table) recalculates a chosen output for a grid of input values, all at once. It is the standard tool for a sensitivity matrix (e.g. share price vs WACC and terminal growth).

- **One-variable data table:** list the input values down a column (or across a row). Put the output formula one cell up and to the right (for a column input) referencing the model output. In the dialog, set **Column input cell** to the model's input cell.
- **Two-variable data table:** input values run down the left column *and* across the top row; the output formula sits in the top-left corner cell of the grid. Set both **Row input cell** and **Column input cell**.

Worked Example 7 — Two-variable data table (DCF share price vs WACC & g)

Say your DCF outputs a per-share value in B2, driven by WACC in B3 and terminal growth g in B4.

1. Pick a blank area. In the **top-left corner cell** of the grid, type =B2 (a link to the output — this cell must reference the result).
2. Down the left column below that corner, list WACC values: 10%, 11%, 12%, 13%.
3. Across the top row to the right of the corner, list g values: 3%, 4%, 5%.
4. Select the whole rectangle (corner + row headers + column headers + empty body).
5. Data → What-If Analysis → Data Table. Set **Row input cell = B4** (g, since g runs across the row) and **Column input cell = B3** (WACC, since it runs down the column).
6. Excel fills every cell with the recomputed share price for that WACC/g pair. It writes an array {=TABLE(B4,B3)} — you cannot edit individual result cells.

Read it like a football-field: the price rises as WACC falls and as g rises. This grid is the single most-used exhibit in an equity research valuation.

Common trap: Getting the row/column input cells backwards, or forgetting that the output-link cell goes in the *top-left corner*. Also: data tables recalculate on every workbook calc and can make large models crawl — set calculation to "Automatic Except for Data Tables" (Formulas → Calculation Options) if the model is slow.

8.3 Scenario Manager

Data → What-If Analysis → Scenario Manager stores named sets of input values (e.g. "Recession", "Base", "Boom") and swaps them in on demand, and can produce a summary report. It is handy for a handful of discrete scenarios but less flexible than a CHOOSE/INDEX toggle for a live, driver-based model; most modelers prefer the toggle approach for its transparency.

8.4 Sensitivity vs scenario — the distinction

Aspect	Sensitivity analysis	Scenario analysis
Varies	One or two inputs at a time, holding all else constant	A coherent set of inputs together (a "story")
Tool	Data table	Toggle / Scenario Manager
Answers	"How sensitive is value to WACC?"	"What happens in a recession?"
Output	A grid / tornado chart	A set of full model outputs per case

8.5 Goal Seek

Goal Seek (Data → What-If Analysis → Goal Seek) answers the inverse question: "what input value makes this output hit a target?" It solves for *one* input to reach *one* target. You specify three things: **Set cell** (the output/formula cell), **To value** (the target number, a hardcode), and **By changing cell** (the single input to solve for — must be a hardcoded value, not a formula).

Worked Example 7A — Goal Seek uses

- **DCF:** "What WACC makes the DCF value equal the current share price?" Set cell = per-share value, To value = market price, By changing cell = WACC input. The answer is the market-implied cost of capital.
- **LBO:** "What entry multiple gives exactly a 25% IRR?" Set cell = IRR, To value = 25%, By changing cell = entry EV/EBITDA. This finds the maximum price the sponsor can pay.
- **M&A:** "How much synergy makes a dilutive deal breakeven?" Set cell = accretion/dilution %, To value = 0%, By changing cell = synergy input. That is breakeven synergies (Section 14).

Goal Seek *overwrites* the input cell with the solved value — note the original first, or work on a copy. It also needs a numeric, continuous relationship; it can fail on step functions or when no solution exists.

8.6 Solver

Solver (an add-in: File → Options → Add-ins → Excel Add-ins → Solver) is Goal Seek's heavyweight cousin. It optimises an objective (maximise, minimise, or hit a value) by changing *multiple* input cells *subject to constraints*. Use it for genuine optimisation: maximise IRR subject to a leverage cap and a minimum DSCR; minimise funding cost across debt tranches subject to a total-debt limit; or portfolio weight optimisation. For everyday three-statement work Goal Seek is usually enough — Solver is reserved for constrained multi-variable problems.

Tool	Changes	Solves for	Constraints?
Goal Seek	1 input	1 output = target	No
Solver	Many inputs	Max / min / target of an objective	Yes
Data table	1–2 inputs (grid)	Output across every combination	No
Scenario Manager	Many inputs (named sets)	Output per stored scenario	No

Common trap: Pointing Goal Seek's "By changing cell" at a formula cell — it only works on a hardcoded input. Also forgetting that Goal Seek permanently changes that input; if you Goal-Seek WACC you have now moved your WACC assumption, which cascades through the whole model.

9. Modeling Best Practices & Conventions

9.1 Colour conventions

The near-universal convention, understood by every banker and analyst:

Colour	Means
Blue font	Hardcoded input / assumption (a number you type)
Black font	Formula / calculation within the same sheet
Green font	Link to another worksheet (same workbook)
Red font	Link to another workbook / external file (fragile — flag it)

This lets anyone open your model and instantly see what is an assumption they can change (blue) versus a calculation they should not overwrite (black).

9.2 The cardinal rules

- **No hardcoding inside formulas.** Never write `=Revenue*1.25` to apply a tax rate — put 25% in a labelled input cell and reference it. Hardcodes buried in formulas are invisible and un-auditable.
- **One formula across a row.** A forecast row should contain the *same* formula copied across every period. If column F differs from column E, that is a red flag (either an error or a deliberate, documented exception).
- **Separate inputs, calculations, outputs.** Ideally on different tabs or clearly boxed. A user should be able to change any blue cell without hunting.
- **Consistent sign convention.** Decide up front: are expenses negative or positive? Model consistently. Most models keep the P&L with positive revenue and positive expenses subtracted; the cash flow uses signed conventions (uses of cash negative).
- **Build error checks.** Balance check, cash-flow-to-balance-sheet cash tie, sum-of-parts checks, and a master "checks" cell that is TRUE only if all sub-checks pass.
- **No naked constants for time.** Use one units/period assumption, not 365 typed in ten places.
- **Document.** A comments column, an assumptions log, and a version/change history. Row labels should be self-explanatory.
- **Keep it simple & transparent.** A model others cannot follow is a liability. Favour clear structure over clever mega-formulas.

9.3 Formatting discipline

- Units stated explicitly (₹ crore) and consistent throughout.
- Historicals vs forecasts visually distinguished.
- Negative numbers in parentheses or red, consistently.
- Freeze panes so labels stay visible; group/outline schedules.
- No merged cells in calculation ranges (they break navigation and formulas).

Common trap: Hardcoding a growth rate or tax rate directly into a formula "just this once". Six months later nobody — including you — can find why the forecast behaves oddly. Every number that could change belongs in a blue input cell.

Key takeaways: Blue = input, black = formula, green = cross-sheet link. No hardcodes in formulas. One formula across the row. Separate inputs/calcs/outputs. Build checks. Document. These conventions are how interviewers judge whether you have real modeling hygiene.

10. Key Excel Functions — Syntax & When to Use

Quote these with correct syntax; interviewers do ask. Excel arguments are comma-separated in the English/US locale (Indian Excel installs typically use the comma too).

10.1 Lookups

INDEX + MATCH (the professional's lookup)

`INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`

MATCH returns the *position* of the lookup value; INDEX returns the value at that position. match_type 0 = exact match. Two-way: `INDEX(matrix, MATCH(row_val, rows, 0), MATCH(col_val, cols, 0))`.

VLOOKUP

`VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

Searches the *first* column of table_array; returns the value in column col_index_num. Use FALSE (or 0) for exact match. Cannot look to the left of the key column.

XLOOKUP (modern; Excel 365 / 2021+)

`XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

Looks up in any direction, defaults to exact match, and has a built-in "if not found" argument. Replaces both VLOOKUP and HLOOKUP.

Worked Example 8 — INDEX/MATCH lookup

You have company names in A2:A50 and their EBITDA margins in D2:D50. You want the margin for "Tata Motors" typed in G1:

`=INDEX(D2:D50, MATCH(G1, A2:A50, 0))`

MATCH finds "Tata Motors" at, say, position 12; INDEX returns the 12th value of D2:D50. If the name might be missing, wrap it: `=IFERROR(INDEX(D2:D50, MATCH(G1, A2:A50, 0)), "Not found")`.

Why not VLOOKUP? If EBITDA margin were in column D and the name in column A, VLOOKUP works only because A is left of D. But if you needed to return a value in column A while looking up by column D, VLOOKUP cannot look left — INDEX/MATCH can. INDEX/MATCH also does not break when columns are inserted (VLOOKUP's hardcoded col_index_num shifts).

10.2 Conditional aggregation

SUMIFS / COUNTIFS / AVERAGEIFS

`SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)`

`COUNTIFS(criteria_range1, criteria1, ...)`

`AVERAGEIFS(average_range, criteria_range1, criteria1, ...)`

Note the argument order: SUMIFS puts the sum_range *first*; the older SUMIF puts it last: `SUMIF(range, criteria, [sum_range])`. Criteria can use operators in quotes, e.g. `">1000"`, `"<="&B1`.

Use SUMIFS to total revenue for a segment, a region, or a date range from a flat data dump — the workhorse of building drivers from raw transaction or trial-balance data.

10.3 Logic & error handling

IF / nested IF / IFS

IF(logical_test, value_if_true, value_if_false)

IFS(test1, val1, test2, val2, ...) (365/2019+) avoids deep nesting.

IFERROR

IFERROR(value, value_if_error)

Returns value_if_error only if value evaluates to an error (#DIV/0!, #N/A, #REF!, ...). Cleaner than IF(ISERROR(...),...). Do not over-use — it can hide genuine problems.

10.4 Selection & dynamic references

CHOOSE

CHOOSE(index_num, value1, value2, ...) — returns the value at position index_num. Ideal for scenario toggles.

OFFSET (volatile — use sparingly)

OFFSET(reference, rows, cols, [height], [width]) — returns a range shifted from a reference. Useful for rolling ranges but recalculates on every change (slows big models).

INDIRECT (volatile)

INDIRECT(ref_text, [a1]) — turns a text string into a cell/range reference, e.g. build a sheet name dynamically. Powerful but volatile and hard to audit; prefer INDEX where possible.

10.5 Time value of money

NPV / XNPV

NPV(rate, value1, value2, ...) — discounts a series of *equally spaced* cash flows, assuming the first cash flow is one period away.

XNPV(rate, values, dates) — discounts cash flows on their actual dates (handles irregular timing).

Trap: NPV assumes the first value occurs at the end of period 1. A time-0 cash flow must be added *outside* NPV: =C0 + NPV(rate, C1:Cn).

IRR / XIRR / MIRR

IRR(values, [guess]) — rate where NPV = 0 for equally spaced flows (first value usually the negative outlay).

XIRR(values, dates, [guess]) — IRR for cash flows on actual dates (the correct tool for real, irregularly timed flows).

MIRR(values, finance_rate, reinvest_rate) — reinvests interim flows at a specified rate.

PMT (loan/annuity payment)

PMT(rate, nper, pv, [fv], [type])

rate is *per period* (divide annual by 12 for monthly). Returns a negative number (a payment out). type = 0 (end of period, default) or 1 (beginning).

Worked Example 9 — IRR vs XIRR

A PE-style cash flow: invest ₹100 cr today, receive ₹0, ₹0, ₹0, ₹0, ₹250 cr in year 5.

Flows in B1:B6 = -100, 0, 0, 0, 0, 250. Annual, equally spaced:

=IRR(B1:B6) → about 20.1% (since $250/100 = 2.5\times$ over 5 years, $IRR = 2.5^{(1/5)} - 1 \approx 20.1\%$).

Now the same cash flows on real dates in A1:A6 (e.g. deal closes 15-Apr-2026, exit 30-Jun-2031). Use:

=XIRR(B1:B6, A1:A6) — this accounts for the exact 5.2-year holding period, giving a slightly different (more accurate) rate. XIRR is what PE and real-estate models use because deal cash flows never land on tidy annual boundaries.

EMI check with PMT: a ₹50 lakh home loan at 9% p.a. for 20 years: =PMT(9%/12, 20*12, 5000000) → about -₹44,986/month.

IPMT / PPMT (splitting a loan payment)

IPMT(rate, per, nper, pv, [fv], [type]) — the *interest* portion of payment number per.

PPMT(rate, per, nper, pv, [fv], [type]) — the *principal* portion of that same payment.

IPMT + PPMT = PMT for every period. Use them to build an amortisation schedule (interest falls, principal rises over the loan's life). Essential for a debt schedule with an amortising term loan.

RATE (solve for the implied rate)

RATE(nper, pmt, pv, [fv], [type], [guess]) — returns the periodic interest rate given the payment stream. Multiply by periods/year to annualise. Useful for backing out the effective rate on a lease or structured loan.

EFFECT / NOMINAL (rate conversion)

EFFECT(nominal_rate, npery) — converts a nominal annual rate compounded npery times into the effective annual rate (EAR).

NOMINAL(effect_rate, npery) — the reverse.

Example: =EFFECT(12%, 12) → 12.68% EAR for 12% nominal compounded monthly. Lets you compare loans quoted on different compounding bases apples-to-apples.

MIRR (modified IRR)

MIRR(values, finance_rate, reinvest_rate) — reinvests interim positive flows at reinvest_rate and finances negatives at finance_rate.

Fixes two IRR flaws: the unrealistic assumption that interim cash is reinvested at the IRR itself, and the multiple-IRR problem when cash flows change sign more than once.

10.6 Dynamic arrays (Excel 365)

Modern Excel "spills" a whole array from one formula. These are increasingly expected and worth quoting.

FILTER / SORT / UNIQUE

FILTER(array, include, [if_empty]) — returns only rows meeting a condition, e.g.

=FILTER(A2:D50, C2:C50="Auto") pulls every auto-segment row.

SORT(array, [sort_index], [sort_order], [by_col]) — sorts an array in a formula.

UNIQUE(array) — the distinct values in a range (great for building a driver list from a data dump).

SEQUENCE(rows, [cols], [start], [step]) — generates a number series (e.g. forecast period numbers).

These "spill" into neighbouring cells and update live. Reference the whole spilled range with the # operator, e.g. =SUM(F2#).

10.7 LET and LAMBDA (concept)

LET — name intermediate results inside one formula

LET(name1, value1, [name2, value2, ...], calculation)

Example: =LET(rev, A1*A2, marg, 0.2, rev*marg). Improves readability and speed (each named part is computed once).

LAMBDA — build a reusable custom function with no VBA

LAMBDA(parameter1, ..., calculation), saved via Name Manager to become a callable function.

Conceptually lets you define, say, a DEPRECIATE() or GORDONTV() function once and reuse it everywhere. Powerful but newer; know it exists and that it replaces repetitive VBA UDFs.

10.8 Text & date functions**Text**

LEFT / **RIGHT** / **MID**(text, num), **LEN**(text), **TRIM**(text) (strip stray spaces), & or **CONCAT** / **TEXTJOIN**(delim, ignore_empty, ...) to join, **TEXT**(value, format) to format a number as text (e.g. =TEXT(A1, "₹#,##0")).

TRIM and **CLEAN** are lifesavers when importing broker or exchange data with hidden spaces that break lookups.

Date

EOMONTH(start, months) — last day of a month N months out (roll monthly model periods).

EDATE(start, months) — same date N months out.

YEARFRAC(start, end, [basis]) — the fraction of a year between two dates (day-count for accrued interest / XIRR-style timing).

DATE / **YEAR** / **MONTH** / **DAY**, **TODAY**() (volatile).

10.9 Power Query (concept)

Power Query (Data → Get & Transform) is Excel's ETL tool: it connects to files, folders, databases, or the web, then *records a repeatable sequence of cleaning steps* (remove columns, unpivot, merge, filter, change types). When the source refreshes, you click Refresh and the whole cleaned table rebuilds — no re-doing the manual clean. It is the professional way to feed a model from messy trial-balance exports, trade blotters, or multiple monthly files, and it keeps raw data out of your calculation sheets. Know it as the tool that automates data prep; it does not do the modeling itself.

10.10 Data & presentation tools

- **Pivot tables** (Insert → PivotTable): summarise large flat datasets by dragging fields into Rows/Columns/Values/Filters — indispensable for slicing trial-balance or trade-blotter data before it enters a model. Refresh after source changes.
- **Conditional formatting** (Home → Conditional Formatting): colour cells by rule — flag the balance check red when non-zero, highlight covenant breaches, build heat-maps on sensitivity tables.
- **Named ranges** (Formulas → Name Manager / Ctrl+F3): give a cell a name like TaxRate so formulas read =EBT*TaxRate. Improves readability; use judiciously (too many names become their own maintenance burden).
- **Data validation** (Data → Data Validation): restrict an input cell to a list (e.g. scenario selector 1/2/3) to prevent bad entries.

10.11 Function reference summary

Function	Syntax (key args)	Use it when...
INDEX/MATCH	INDEX(rng, MATCH(val, rng, 0))	Flexible two-way lookup; look left; robust to column inserts
VLOOKUP	VLOOKUP(val, tbl, col, FALSE)	Quick left-to-right lookup on a fixed table
XLOOKUP	XLOOKUP(val, look, return, [nf])	Modern replacement; any direction; built-in not-found
SUMIFS	SUMIFS(sum, crit_rng, crit, ...)	Total values meeting multiple conditions
COUNTIFS / AVERAGEIFS	Same pattern	Count / average with conditions
IF / IFS	IF(test, a, b)	Branching logic; IFS avoids deep nesting
IFERROR	IFERROR(val, alt)	Trap #DIV/o!, #N/A gracefully
CHOOSE	CHOOSE(n, v1, v2, ...)	Scenario toggles; pick the n-th option
OFFSET	OFFSET(ref, r, c, [h], [w])	Rolling/dynamic ranges (volatile — sparingly)
INDIRECT	INDIRECT(text)	Reference built from text/sheet name (volatile)
NPV / XNPV	NPV(rate, flows) / XNPV(rate, flows, dates)	Discount future cash flows (regular / dated)
IRR / XIRR	IRR(flows) / XIRR(flows, dates)	Return that sets NPV = 0 (regular / dated)
PMT	PMT(rate, nper, pv)	Loan/annuity payment (EMI)
EOMONTH / EDATE	EOMONTH(start, months)	Roll dates for monthly models
ROUND	ROUND(num, digits)	Control precision (never ROUND inside live links carelessly)
IPMT / PPMT	IPMT(rate, per, nper, pv)	Interest / principal split of an amortising payment
RATE	RATE(nper, pmt, pv)	Back out the implied periodic interest rate
EFFECT	EFFECT(nominal, npery)	Nominal → effective annual rate (compare loans)
MIRR	MIRR(vals, fin, reinv)	IRR with realistic reinvestment; avoids multiple-IRR
FILTER / UNIQUE / SORT	FILTER(arr, include)	Dynamic-array slice / distinct / sort (365)
LET / LAMBDA	LET(name, val, calc)	Name sub-results; build reusable functions (365)
TEXTJOIN / TEXT	TEXT(val, "☛#,##0")	Join / format text and labels
YEARFRAC	YEARFRAC(start, end)	Day-count fraction between dates

11. Keyboard Shortcuts Reference

Fluency with keyboard shortcuts is a visible signal of experience in a modeling test. Learn these cold. (Windows Excel; the interviewer's machine will be Windows.)

Shortcut	Action
F2	Edit the active cell (enter edit mode)
F4	Toggle absolute/relative reference (\$) while editing; repeat last action otherwise
F9	Recalculate all; on a selected formula part, evaluate just that part
Ctrl + arrow	Jump to the edge of a data region
Ctrl + Shift + arrow	Select to the edge of a data region
Ctrl + Space / Shift + Space	Select entire column / entire row
Ctrl + C / X / V	Copy / Cut / Paste
Ctrl + Alt + V	Paste Special (then V = values, F = formulas, T = formats)
Alt + =	AutoSum
Ctrl + D / Ctrl + R	Fill down / Fill right (copy formula across the row — enforces "one formula")
Ctrl + ;	Insert today's date (static)
Ctrl + 1	Format Cells dialog
Ctrl + Shift + 4 / 5 / 1	Currency / Percent / Number (comma) format
Ctrl + Page Up / Page Down	Move between worksheet tabs
Ctrl + F3 / F3	Name Manager / paste a name
Ctrl + [(open bracket)	Trace precedents (go to cells this formula depends on)
Ctrl + ` (grave)	Toggle show formulas / show values
Alt + Enter	New line within a cell
Ctrl + Shift + L	Toggle AutoFilter
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + PgDn then Alt+E+S	Common paste-special muscle memory (legacy)
Alt (then ribbon letters)	Keyboard access to every ribbon command (e.g. Alt+A+W+T = Data Table)
F5 / Ctrl+G → Special	Go To Special (select constants, formulas, blanks — great for auditing hardcodes)

Key takeaways: F2, F4, Ctrl+arrows, Ctrl+Shift+arrows, Alt+=, Ctrl+D/R, Ctrl+Alt+V, and the Alt-ribbon chords are the highest-leverage. Using **Go To Special → Constants** to hunt down every hardcoded number in the calculation area is a pro auditing move worth mentioning in interviews.

12. DCF Model Basics

A discounted cash flow model values a company as the present value of its future free cash flows. It sits on top of the three-statement model, which supplies the cash flows.

Unlevered DCF (enterprise value)

$FCFF = EBIT \times (1 - \text{tax}) + D\&A - \text{Capex} - \Delta \text{Net working capital}$

$\text{Enterprise value} = \sum [FCFF_t / (1 + WACC)^t] + \text{Terminal value} / (1 + WACC)^n$

$\text{Terminal value (Gordon growth)} = FCFF_n \times (1 + g) / (WACC - g)$

$\text{Equity value} = \text{Enterprise value} - \text{Net debt}; \text{ Per share} = \text{Equity value} / \text{shares}$

WACC = weighted average cost of capital; g = perpetual growth rate (< long-run GDP growth). Terminal value often exceeds 60–75% of total — sanity-check it with an exit-multiple cross-check.

In Excel: build FCFF as a row off the three-statement model; discount with explicit period factors $=1/(1+WACC)^t$ or XNPV for dated flows; compute terminal value both ways (Gordon growth and exit EV/EBITDA multiple) and compare; then run the two-variable data table (WACC $\times$ g) from Worked Example 7.

Common trap: Using NPV for the mid-year convention or for the terminal value without care. NPV assumes year-end flows one period out; if you discount terminal value, discount it by the *same* n as the final explicit year, and remember NPV excludes any time-0 cash flow.

13. LBO Model Basics

A leveraged buyout model evaluates buying a company using a large slug of debt, running it for ~3–5 years, and selling it. Returns come from (a) paying down debt with the company's cash (deleveraging), (b) EBITDA growth, and (c) multiple expansion. Private equity firms live in this model.

13.1 Sources & Uses

The transaction structure starts with a Sources & Uses table — where the money comes from and where it goes. The two sides must equal.

Sources & Uses

Uses: Purchase of equity (Enterprise value = $EV/EBITDA \times EBITDA$) + refinance existing debt + transaction fees

Sources: New debt (term loans + bonds) + sponsor equity (the plug) + cash on hand

Sponsor equity = Total uses – Total debt – Cash used

Sponsor equity is the plug on the sources side: whatever debt and cash cannot fund, the PE fund writes as an equity cheque.

13.2 Debt paydown & returns

Each year the company's free cash flow (after interest and mandatory amortisation) sweeps down the debt. Lower debt → lower interest → more cash to sweep — a virtuous deleveraging cycle. At exit, you sell at an assumed EV/EBITDA multiple, repay remaining debt, and the residual is the equity proceeds.

LBO returns

Exit enterprise value = Exit EV/EBITDA multiple × Exit-year EBITDA

Exit equity value = Exit EV – Net debt at exit

MOIC (money multiple) = Exit equity value / Initial sponsor equity

IRR = $MOIC^{(1 / \text{holding years})} - 1$ (for a single-in, single-out cash flow)

MOIC = Multiple On Invested Capital (also "cash-on-cash"). IRR is time-weighted; compute precisely with XIRR on dated equity flows.

Worked Example 10 — Quick LBO returns

Buy a company for 8.0× EBITDA of ₹500 cr → EV ₹4,000 cr. Fund with ₹2,400 cr debt (4.8×) and ₹1,600 cr sponsor equity. Over 5 years EBITDA grows to ₹700 cr and net debt is paid down to ₹1,000 cr. Exit at 8.0× (no multiple expansion).

- Exit EV = $8.0 \times 700 = ₹5,600$ cr.
- Exit equity = $5,600 - 1,000 = ₹4,600$ cr.
- MOIC = $4,600 / 1,600 = 2.875\times$.
- IRR $\approx 2.875^{(1/5)} - 1 = \sim 23.5\%$.

Notice all three levers at work: EBITDA grew (500→700), debt fell (2,400→1,000 net), even with a flat exit multiple. This is the intuition PE interviewers want.

13.3 The debt tranches (capital structure)

LBO debt is layered. Each tranche sits at a different point in the risk/return and repayment hierarchy — senior/secured debt is cheapest and gets repaid first; subordinated debt is dearer and repaid last. Know the

stack:

Tranche	Rank	Cost / features	Amortisation
Revolver (RCF)	Senior secured	Cheapest; undrawn line for working-capital swings; commitment fee on undrawn portion	Drawn/repaid flexibly; the plug
Term Loan A (TLA)	Senior secured	Bank-held, floating rate; amortising	Straight-line (e.g. 5–10%/yr) + sweep
Term Loan B (TLB)	Senior secured	Institutional (funds/CLOs), floating; longer tenor	Tiny mandatory (~1%/yr) "bullet"; cash sweep
High-yield bonds	Senior/sub unsecured	Fixed coupon, higher rate; often non-call for a period	Bullet at maturity (no amortisation)
Mezzanine	Subordinated	Highest cost; may carry PIK interest and equity warrants	Bullet; repaid last

Leverage is quoted in turns of EBITDA (e.g. "4.8× total, 3.0× senior"). More debt magnifies equity returns but raises interest and default risk — the central LBO trade-off. **PIK** (payment-in-kind) interest accrues onto the principal rather than being paid in cash, preserving near-term cash flow at the cost of a larger balance later.

13.4 The cash sweep & debt paydown

After paying mandatory amortisation and interest, the LBO directs surplus cash to prepay debt — the **cash sweep** — in strict seniority order (revolver, then TLA, then TLB, then bonds). A "100% cash sweep" applies all free cash; sponsors sometimes model a partial sweep to retain a cash cushion.

Cash sweep waterfall (per year)

Cash available for sweep = Beginning cash + CFO – Capex – Mandatory amortisation – Min. cash

Sweep tranche 1 (revolver) = min(Cash available, Revolver balance)

Remaining cash → next tranche, and so on down the seniority ladder

Ending debt (each tranche) = Beginning – Mandatory – Sweep prepayment

Lower debt → lower interest next year → more cash to sweep: the deleveraging flywheel. The sweep interacts with interest, creating the same circularity as the three-statement model.

13.5 The returns bridge (value-creation attribution)

A returns (or value-creation) bridge decomposes the equity gain into its three sources, so you can see *where* the money came from. From the Worked Example 10 deal (entry equity ₹1,600 cr → exit equity ₹4,600 cr, a ₹3,000 cr gain):

Returns bridge components

EBITDA growth = $\Delta \text{EBITDA} \times \text{entry multiple} = (700 - 500) \times 8.0 = \text{+₹1,600 cr}$

Multiple expansion = $\Delta \text{Multiple} \times \text{exit EBITDA} = (8.0 - 8.0) \times 700 = \text{₹0}$

Debt paydown / FCF = $\Delta \text{Net debt} = (2,400 - 1,000) = \text{+₹1,400 cr}$

Total equity gain = $1,600 + 0 + 1,400 = \text{₹3,000 cr} \checkmark$

A "good" LBO earns most of its return from operational EBITDA growth and deleveraging, not from betting on multiple expansion (which the sponsor does not control).

13.6 Sensitivity in an LBO

The standard LBO output is a data-table matrix of **IRR (and MOIC)** against the levers that matter most:

- **Entry multiple × Exit multiple** — the single most-shown LBO sensitivity; shows how much return depends on multiple expansion vs paying up.
- **Leverage (entry Debt/EBITDA) × Exit multiple** — more leverage lifts IRR when the deal works but craters it if the exit multiple compresses.
- **EBITDA growth (or exit EBITDA) × Exit multiple.**

Build these with the two-variable data table from Worked Example 7 (output-link in the corner, one lever down the column, one across the row). Interviewers love: "if you pay one turn more at entry but exit at the same multiple, what happens to IRR?" — it falls, because you invested more equity for the same exit.

14. M&A / Accretion–Dilution

An M&A model tests whether an acquisition increases (accretive) or decreases (dilutive) the acquirer's earnings per share. A quick accretion/dilution check is a favourite interview drill.

Accretion / Dilution logic

Pro-forma net income = Acquirer NI + Target NI + after-tax synergies – after-tax new interest (on acquisition debt) – foregone interest on cash used

Pro-forma shares = Acquirer shares + new shares issued to fund the deal

Pro-forma EPS = Pro-forma net income / Pro-forma shares

Accretive if Pro-forma EPS > Acquirer standalone EPS; dilutive if lower.

Rule of thumb (all-stock deal): if the acquirer's P/E is *higher* than the target's P/E, the deal is accretive; if lower, dilutive. Cash/debt-funded deals turn on whether the target's earnings yield (E/P) exceeds the after-tax cost of the debt/cash used.

Worked Example 11 — Accretion/dilution quick check

Acquirer: net income ₹1,000 cr, 100 cr shares → EPS ₹10; trades at P/E 20 (price ₹200). Target: net income ₹200 cr, offered ₹4,000 cr, funded entirely by issuing new acquirer shares at ₹200.

- New shares issued = $4,000 / 200 = 20$ cr shares.
- Pro-forma NI = $1,000 + 200 = ₹1,200$ cr (ignore synergies).
- Pro-forma shares = $100 + 20 = 120$ cr.
- Pro-forma EPS = $1,200 / 120 = ₹10.00$ → exactly neutral.

Cross-check with the P/E rule: target implied P/E = $4,000 / 200 = 20$, equal to the acquirer's P/E of 20 → neutral. If the acquirer instead paid only ₹3,000 cr (target P/E 15 < 20) the deal would be accretive; paying ₹5,000 cr (P/E 25 > 20) would be dilutive.

14.1 Purchase price & funding mix

The **offer price** is usually quoted per share at a premium (20–40%) to the target's undisturbed price; multiply by shares (plus in-the-money options, the treasury-stock method) for equity purchase price. Add net debt assumed for the enterprise value. The acquirer then chooses a **funding mix** — cash on hand, new debt, and/or new equity (stock). Each source has a cost: cash foregoes interest income, debt costs after-tax interest, stock dilutes by adding shares. The mix drives accretion/dilution.

14.2 Purchase accounting & goodwill creation

Under acquisition accounting (Ind AS 103 / IFRS 3), the acquirer records the target's assets and liabilities at **fair value**, not book value. This "step-up" often writes up PP&E and recognises previously unrecorded **intangibles** (brands, customer relationships). Whatever the purchase price exceeds the fair value of net identifiable assets becomes **goodwill** — the residual plug on the pro-forma balance sheet.

Goodwill created

Goodwill = Equity purchase price – Fair value of net identifiable assets acquired

Fair value of net identifiable assets = FV assets (incl. asset step-ups & new intangibles) – FV liabilities (incl. deferred tax liability on step-ups)

Under Ind AS/IFRS, goodwill is *not* amortised — it is tested annually for impairment. Identifiable intangibles with finite lives *are* amortised, which creates incremental (often tax-deductible or non-deductible) expense that hits pro-forma EPS. A DTL arises because the fair-value step-up is booked for accounting but not for tax.

Worked Example 11A — Goodwill calculation

Acquirer pays ₹4,000 cr equity for a target whose book equity is ₹1,500 cr. Fair-value review steps PP&E up by ₹400 cr and identifies a ₹600 cr brand intangible; a deferred tax liability of ₹250 cr arises on the step-ups.

- Fair value of net identifiable assets = $1,500 + 400 + 600 - 250 = ₹2,250$ cr.
- Goodwill = $4,000 - 2,250 = ₹1,750$ cr, recorded as an asset on the pro-forma balance sheet.

The ₹600 cr brand, if amortised over 10 years, adds ₹60 cr/yr of amortisation that depresses pro-forma net income (and EPS) even though it is a non-cash charge.

14.3 Synergies

Synergies are the extra value from combining the two firms and are what justify the control premium. Two kinds: **cost synergies** (headcount, procurement scale, closing duplicate facilities — the credible, modellable kind) and **revenue synergies** (cross-selling, wider distribution — real but softer, so analysts haircut them). Model synergies net of the one-time **cost to achieve** (restructuring, integration) and phase them in over 2–3 years. In accretion/dilution, add *after-tax* synergies to pro-forma net income.

14.4 Breakeven synergies

For a dilutive deal, the key question is: "how much synergy makes it breakeven (EPS-neutral)?" Solve for the pre-tax synergy that lifts pro-forma EPS back to the standalone EPS — either algebraically or with Goal Seek (Set pro-forma EPS = standalone EPS, By changing = synergy cell).

Breakeven synergies

Required after-tax synergy = (Standalone EPS × Pro-forma shares) – Pro-forma NI before synergies

Required pre-tax synergy = After-tax synergy / (1 – tax rate)

Worked Example 11B — Breakeven synergies

Acquirer standalone: NI ₹1,000 cr, 100 cr shares → EPS ₹10. It buys a target for ₹3,000 cr of new debt at 8% pre-tax; target NI ₹150 cr; no new shares; tax 25%.

- After-tax new interest = $3,000 \times 8\% \times (1 - 25\%) = ₹180$ cr.
- Pro-forma NI before synergies = $1,000 + 150 - 180 = ₹970$ cr; shares stay 100 cr → EPS ₹9.70 → **dilutive by ₹0.30**.
- Required after-tax synergy = $(10 \times 100) - 970 = ₹30$ cr.
- Required pre-tax synergy = $30 / (1 - 0.25) = ₹40$ cr. If management can credibly deliver more than ₹40 cr of pre-tax synergies, the deal turns accretive.

14.5 Building the pro-forma combination

A full merger model combines the two three-statement models line by line: add the income statements (then layer synergies and the incremental D&A from intangibles and new interest), and combine the balance sheets with the purchase-accounting adjustments (eliminate target equity, add goodwill and step-ups, add acquisition debt, reduce cash used, add new shares to equity). The pro-forma balance sheet must still balance — goodwill is precisely the plug that makes it so.

15. Types of Models — Comparison

Model	Purpose	Core output	Who uses it
Three-statement	Project P&L, BS, CF in an integrated way	Forecast financials; the base for all others	Everyone (ER, IB, FP&A, PC)
DCF	Intrinsic valuation from future cash flows	Enterprise/equity value, per-share value	Equity research, IB valuation
LBO	Test a debt-funded buyout's returns	IRR, MOIC, max purchase price	Private equity, IB leveraged finance
M&A / accretion-dilution	Test EPS impact of an acquisition	Accretive/dilutive, pro-forma EPS, synergies needed	IB M&A, corp dev
Comparable companies / precedent transactions	Relative valuation via multiples	Valuation range (EV/EBITDA, P/E)	ER, IB
Budget / operating model (FP&A)	Annual plan, rolling forecast, variance	Budget vs actual, driver-based plan	FP&A, corporate finance
Sum-of-the-parts (SOTP)	Value a conglomerate by segment	Segment-level EV summed to group value	ER (e.g. valuing a diversified group)

Key takeaways: The three-statement model is the foundation; DCF, LBO, and M&A models bolt onto it. Know each model's *purpose* and *headline output* — that is what interviewers probe when they ask "what model would you build for X?".

16. Interview Q&A Bank

16.1 Three-statement mechanics

Q1: Walk me through a three-statement model.

A: Start with historical financials and clean them up so they tie out. Build assumption-driven supporting schedules — revenue build, cost build, depreciation, debt & interest, working capital, and an equity roll-forward. Project the income statement down to EBIT. Then project balance-sheet items driven by the schedules: PP&E from the depreciation schedule, working capital from days ratios, debt from the debt schedule, equity from the roll-forward. Build the cash flow statement by starting from net income, adding back non-cash items like D&A, adjusting for changes in working capital (CFO), subtracting capex and adding asset sales (CFI), and adding debt draws and equity issuance less repayments and dividends (CFF). The cash flow's ending cash links into the balance-sheet cash line; interest from the debt/cash balances feeds back into the P&L, creating an intentional circularity resolved by iterative calc or a beginning-balance convention. Finally, the balance-check row (Assets – Liabilities – Equity) must read zero in every year.

Q2: How are the three statements linked?

A: Net income flows from the P&L into the top of the cash flow statement and into retained earnings on the balance sheet. D&A is a P&L expense, a cash-flow add-back, and reduces net PP&E on the balance sheet. Capex is a cash outflow in investing and increases gross PP&E. Changes in working-capital accounts and in debt/equity move between the balance sheet and the cash flow. The single most important link is that the cash flow statement's ending cash becomes the balance-sheet cash. In short, the cash flow statement is the bridge that reconciles the opening and closing balance sheet.

Q3: What makes the balance sheet balance?

A: The accounting identity, Assets = Liabilities + Equity, enforced by double-entry bookkeeping — every transaction hits two places that preserve the identity. In a model it balances automatically when the links are correct: the cash flow statement captures every change in every balance-sheet account (as either a non-cash P&L reversal or the change in an account), so ending cash plus retained earnings move exactly in step with the rest of the balance sheet. If a model doesn't balance, it's a linking error, not an accounting impossibility.

Q4: If you could use only one statement to evaluate a company, which and why?

A: The cash flow statement. It shows the actual cash the business generates, is harder to manipulate than accrual-based earnings, and reveals whether reported profits are converting into cash. That said, in practice you need all three — the cash flow statement is itself derived from the P&L and the change in the balance sheet.

Q5: Depreciation goes up by ₹100 with a 25% tax rate. Walk through all three statements.

A: P&L: pre-tax income falls ₹100, tax falls ₹25, net income falls ₹75. Cash flow: net income – ₹75 but add back ₹100 of D&A → CFO up ₹25, so cash rises ₹25. Balance sheet: net PP&E down ₹100, cash up ₹25 (assets –75); retained earnings down ₹75 (equity –75). Both sides fall ₹75 — it balances. The +₹25 is the depreciation tax shield.

Q6: You buy ₹100 of inventory with cash. Walk through the statements.

A: No P&L impact yet (inventory is capitalised, not expensed until sold). Balance sheet: inventory +₹100, cash –₹100 — assets unchanged in total, so it balances. Cash flow: an increase in inventory is a use of cash, so CFO –₹100, reducing cash ₹100. When the inventory is later sold, COGS hits the P&L and the cycle reverses.

Q7: A company raises ₹500 of debt. Walk through the statements.

A: No immediate P&L effect (interest comes later). Cash flow: +₹500 financing inflow → cash up ₹500. Balance sheet: cash +₹500 (asset), debt +₹500 (liability) — both sides up ₹500, balances. In subsequent periods interest expense hits the P&L, reducing net income and cash.

Q8: Why do we add back depreciation in the cash flow statement?

A: Because it's a non-cash expense. It reduced accounting profit but no cash actually left the business in that period (the cash left earlier, when the asset was bought, and shows up as capex in investing). The indirect method starts from net income, which already subtracted D&A, so we add it back to get to cash generated.

Q9: What's the difference between the direct and indirect cash flow methods?

A: The direct method lists actual cash receipts and payments (cash from customers, cash to suppliers). The indirect method starts from net income and adjusts for non-cash items and changes in working capital. Models almost always use the indirect method because it links cleanly to the P&L and balance sheet and is what most companies report.

16.2 Balancing, plugs, and debugging**Q10: Your model doesn't balance. How do you debug it?**

A: Systematically. First find the earliest year that breaks — the error was introduced there. Note the imbalance's size and sign and Ctrl+F for that number; it often equals a specific line item. Check the cash flow statement first: the usual culprit is a balance-sheet item that changed without a matching cash-flow line. Verify sign conventions — an imbalance of exactly 2× a line item signals a flipped sign. Confirm ending cash links from the cash flow into the balance sheet, and that retained earnings rolls forward correctly (prior RE + net income – dividends). Finally check each schedule ties to its balance-sheet line.

Q11: What is the "plug" in a model?

A: The item that absorbs the residual so the model closes. The legitimate plugs are cash (asset side) and the revolver or short-term debt (liability side): surplus cash sweeps the revolver, a shortfall draws on it. An illegitimate plug is hardcoding a balancing number into an unrelated line to force a balance — that hides an error rather than fixing it.

Q12: How does the revolver work in a model?

A: It's a flexible credit line acting as the financing plug. Compute cash available before the revolver. If it's below the minimum cash policy, draw the shortfall; if above, sweep the surplus to repay the outstanding revolver, but never below zero. Ending cash then sits at or above the minimum. Interest on the revolver feeds back into the P&L, which is the main source of circularity.

Q13: An imbalance is exactly twice a line item. What does that tell you?

A: Almost certainly a sign error — you added something that should have been subtracted (or vice versa), so the item is counted with the wrong sign and the miss is 2× its value. An imbalance equal to one line item usually means a missing link (a balance-sheet change with no matching cash-flow entry).

Q14: How would you check a model for errors generally?

A: Build explicit checks: a balance check ($\text{Assets} - \text{L} - \text{E} = 0$), a tie between cash-flow ending cash and balance-sheet cash, schedule roll-forwards that tie to their statement lines, and a master "all checks pass" cell. Use Go To Special → Constants to hunt hardcodes buried in the calculation area. Trace precedents (Ctrl+[) on suspicious cells. Sanity-check outputs against reality (margins, growth, terminal value share). Confirm one consistent formula across each forecast row. Stress the model with extreme inputs and see if it still balances.

Q15: Where should cash come from on the balance sheet?

A: Always from the cash flow statement's ending cash balance — never a hardcoded. Beginning cash + change in cash (CFO+CFI+CFF) = ending cash, and that cell is what the balance sheet references. Hardcoding cash to force a balance is the classic plugging-cash-wrongly mistake.

16.3 Circular references**Q16: What causes circular references in a model, and how do you handle them?**

A: The classic one runs through interest: interest depends on the debt and cash balances → interest hits net income → net income drives cash flow → cash flow changes the cash/revolver balances → which change interest. It's a legitimate, intentional loop. Handle it by enabling iterative calculation (File → Options → Formulas), which lets Excel iterate to convergence, and by building a circularity switch — a 1/0 cell that forces interest to zero to break the loop and clear any propagated error. Alternatively, charge interest on the opening balance to avoid the circularity entirely, or compute once and paste values.

Q17: What is a circularity switch and why build one?

A: A single input cell (1 = on, 0 = off) wrapped around the interest calc: `=IF(switch=1, rate*avg_debt, 0)`. If an error ever races around the circular loop and freezes the model, flip the switch to 0 — interest becomes zero, the loop opens, the error clears, and the model recalculates cleanly; then flip back to 1. It's a circuit breaker for the fragility of iterative calculation.

Q18: What are the downsides of iterative calculation?

A: Fragility and speed. If any cell in the loop returns an error, it propagates around the circle and can freeze the whole model, and toggling iteration may not clear it without a switch. Large iterative models also recalc slowly. And results depend on convergence settings — set enough iterations and a small max-change threshold.

Q19: How can you avoid circularity altogether?

A: Charge interest on the beginning debt balance instead of the average — the interest no longer depends on this period's cash generation, so the loop is broken. It costs a little precision but is common in interview-safe and buy-side models. Copy-paste-values on the interest line is another way, but then interest is static until you repeat the paste.

16.4 Excel functions**Q20: INDEX/MATCH vs VLOOKUP — which and why?**

A: I prefer INDEX/MATCH. VLOOKUP can only look to the right of its key column and uses a hardcoded column-index number that silently breaks when you insert or delete columns. INDEX/MATCH looks in any direction, is more robust to structural changes, and is faster on large sheets because it only scans the lookup and return columns. Syntax: `=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`. On Excel 365 I'd use XLOOKUP, which combines the best of both with a built-in not-found argument.

Q21: What does the 0 (or FALSE) in MATCH/VLOOKUP do?

A: It forces an exact match. MATCH's third argument 0 means exact; 1 (or omitted) assumes the data is sorted ascending and finds the largest value ≤ lookup; -1 assumes descending. VLOOKUP's last argument FALSE (0) is exact match; TRUE (1) is approximate and requires a sorted first column. In modeling you almost always want exact match, so use 0/FALSE.

Q22: Explain SUMIFS and how its argument order differs from SUMIF.

A: SUMIFS(sum_range, criteria_range1, criteria1, ...) totals the sum_range for rows meeting all the criteria; it supports multiple conditions. Crucially the sum_range comes *first*. The older SUMIF(range, criteria, [sum_range]) puts the sum_range *last*. Mixing them up is a common error. Criteria can embed operators, e.g. ">="&B1.

Q23: When would you use IFERROR, and what's the risk?

A: IFERROR(value, value_if_error) traps errors like #DIV/o! or #N/A and returns a fallback — useful for clean lookups or ratios that might divide by zero. The risk is masking genuine problems: if you wrap everything in IFERROR you can hide a broken link that should be investigated. Use it deliberately, not as a blanket.

Q24: NPV in Excel — what's the common mistake?

A: NPV(rate, values) assumes the first value is one period in the future (end of year 1) and that flows are equally spaced. So a time-0 cash flow (like an upfront investment) must be added outside the function: =CF0 + NPV(rate, CF1:CFn). Putting CF0 inside NPV over-discounts everything by one period. For irregular dates use XNPV, which takes explicit dates.

Q25: IRR vs XIRR?

A: IRR assumes equally spaced periods; XIRR takes actual dates and is correct for real cash flows that don't land on annual boundaries — the standard for PE, real estate, and deal cash flows. =XIRR(values, dates, [guess]). Both find the rate that sets NPV to zero. IRR can also mislead with non-conventional cash flows (multiple sign changes give multiple IRRs), where MIRR or NPV is safer.

Q26: What's the difference between OFFSET/INDIRECT and INDEX for dynamic references?

A: OFFSET and INDIRECT are *volatile* — they recalculate on every change anywhere in the workbook, which slows large models, and they're harder to audit (INDIRECT builds references from text, so trace-precedents can't follow them). INDEX is non-volatile and usually achieves the same dynamic behaviour, so prefer INDEX/MATCH where possible; reserve OFFSET/INDIRECT for genuine cases like variable sheet names.

Q27: How do you build a scenario toggle in Excel?

A: Lay out Base/Bull/Bear assumption rows, add a selector cell (1/2/3), and have each driver read the selected case with =CHOOSE(selector, base, bull, bear) or =INDEX(cases_range, selector). Optionally link a Form Control drop-down and use Data Validation to restrict the selector. One cell then flips the entire model.

Q28: Explain a two-variable data table.

A: It recalculates one output across a grid of two inputs. Put a link to the output in the top-left corner cell, list one input's values down the left column and the other's across the top row, select the whole rectangle, then Data → What-If → Data Table and set the Row input cell (for the top-row variable) and Column input cell (for the left-column variable). Excel fills the grid with {=TABLE(. . .)} — a classic WACC × terminal-growth sensitivity for a DCF.

Q29: What is F4 used for in modeling?

A: While editing a formula, F4 cycles a reference through absolute and relative forms — A1 → \$A\$1 → A\$1 → \$A1 → A1. You use it constantly to lock rows/columns before copying a formula across a row or down a column so the right references stay fixed. Outside edit mode, F4 repeats the last action.

Q30: How do you quickly find all hardcoded numbers in a model?

A: Select the calculation area, press F5 (or Ctrl+G) → Special → Constants → Numbers only. Excel selects every cell containing a typed number rather than a formula, so you can spot hardcodes hidden in what should be formula rows. Colour-coding (blue inputs, black formulas) plus this check keeps a model clean.

16.5 Schedules & drivers**Q31: How do you forecast revenue?**

A: Prefer a driver-based build over a blanket growth rate: decompose into volume × price, units × ASP, customers × ARPU, stores × sales-per-store, or (for a bank) average earning assets × net interest margin. This ties the forecast to real business levers and lets scenarios flex each driver independently. Use a simple growth rate only when data or materiality doesn't justify more detail, and always state the assumption.

Q32: How do you build a depreciation schedule?

A: Split it into depreciation on existing assets (depreciate the opening net book value over its remaining life) and depreciation on new capex (each year's additions depreciate over their useful life, starting the year spent). Sum to total depreciation, which hits the P&L and is added back in CFO; roll net PP&E forward as beginning + capex – depreciation. If a full vintage schedule isn't warranted, approximate depreciation as a % of gross block or revenue, calibrated to history.

Q33: How do you forecast working capital?

A: Use activity ratios in days: receivables = $DSO/365 \times \text{revenue}$, inventory = $DIO/365 \times \text{COGS}$, payables = $DPO/365 \times \text{COGS}$. The change in net working capital flows to the cash flow statement — an increase in NWC uses cash. The cash conversion cycle ($DSO + DIO - DPO$) is a useful summary; deteriorating days is a cash-flow warning.

Q34: How does the debt schedule connect to the rest of the model?

A: It rolls each tranche forward (beginning + draws – repayments = ending) and computes interest = rate × balance. Ending debt links to the balance sheet; interest expense links into the P&L below EBIT; draws and repayments hit financing cash flow. Charging interest on the average balance is more accurate but creates circularity; on the beginning balance avoids it.

Q35: How does retained earnings roll forward?

A: Ending retained earnings = beginning RE + net income – dividends (– any buyback charged to RE). It's the link that carries P&L profit onto the balance sheet's equity. A common bug is omitting dividends or referencing the wrong year, which throws the balance off by exactly the dividend or the prior-year profit.

Q36: What drives capex in a forecast?

A: Either a maintenance + growth split, a % of revenue (calibrated to history), or an explicit project pipeline. Capex hits investing cash flow and increases gross PP&E; it should be at least maintenance capex ≈ depreciation to keep the asset base intact, and higher if the business is expanding capacity.

16.6 Valuation & model types**Q37: What is a DCF and how does it relate to the three-statement model?**

A: A DCF values a business as the present value of its future free cash flows plus a discounted terminal value, using WACC as the discount rate. It sits on top of a three-statement model, which produces the free cash flows ($FCFF = EBIT \times (1 - \text{tax}) + D\&A - \text{capex} - \Delta NWC$). Enterprise value = PV of explicit FCFF + PV of terminal value; subtract net debt for equity value, divide by shares for per-share value.

Q38: How do you calculate terminal value?

A: Two ways. Gordon growth (perpetuity): $TV = FCFF_{\text{final}} \times (1+g)/(WACC - g)$, with g below long-run GDP growth. Exit multiple: $TV = \text{exit EV/EBITDA} \times \text{terminal-year EBITDA}$. Best practice is to compute both and cross-check; terminal value often exceeds two-thirds of total value, so it deserves scrutiny.

Q39: Walk me through an LBO.

A: A PE firm buys a company mostly with debt, improves it, and exits in 3–5 years. Set up Sources & Uses: uses = purchase EV + refinanced debt + fees; sources = new debt + sponsor equity (the plug) + cash. Project the three-statement model; each year free cash flow after interest and mandatory amortisation sweeps the debt, deleveraging the company. At exit, apply an EV/EBITDA multiple to exit EBITDA, subtract remaining net debt for exit equity. Returns: $MOIC = \text{exit equity} / \text{initial equity}$; IRR from the dated equity flows. Value comes from deleveraging, EBITDA growth, and multiple expansion.

Q40: What are the three ways an LBO creates value?

A: (1) Debt paydown/deleveraging — using the company's cash flow to reduce debt, so a larger share of a stable enterprise value accrues to equity; (2) EBITDA growth — via revenue growth and margin improvement; (3) multiple expansion — exiting at a higher EV/EBITDA than entry (the least controllable and least relied-upon lever).

Q41: What's the difference between MOIC and IRR?

A: MOIC (money multiple, cash-on-cash) = total cash returned ÷ cash invested — it ignores timing. IRR is the time-weighted annualised return that sets NPV to zero. A 3× MOIC over 3 years is a far better IRR than 3× over 8 years. PE firms quote both because MOIC shows absolute profit and IRR shows speed; a quick fund needs both to be strong.

Q42: Is an all-stock acquisition accretive or dilutive — how do you tell quickly?

A: Compare P/E multiples. If the acquirer's P/E is higher than the price-to-earnings it pays for the target, the deal is accretive to EPS; if lower, dilutive; if equal, neutral. Intuitively, a high-P/E acquirer is issuing "expensive" shares to buy "cheaper" earnings. For cash/debt deals, compare the target's earnings yield to the after-tax cost of the cash or debt used.

Q43: What models would you build to value and pitch an acquisition?

A: A three-statement model for both companies, a DCF and a comparable-companies/precedent-transactions analysis for standalone valuation, an accretion/dilution model to test EPS impact and synergy requirements, and often an LBO to establish a floor value (what a financial sponsor could pay). Together these triangulate a value range and the deal's strategic and financial merits.

Q44: What's the difference between sensitivity and scenario analysis?

A: Sensitivity varies one or two inputs at a time, holding everything else constant, to see how responsive an output is — typically via a data table (e.g. value vs WACC). Scenario analysis changes a coherent bundle of inputs together to represent a story (recession, base, boom) — typically via a toggle or Scenario Manager. Sensitivity answers "how fragile is my answer to this one lever"; scenarios answer "what happens under this whole worldview".

16.7 Best practice & behavioural

Q45: What are your top modeling best practices?

A: Colour-code (blue inputs, black formulas, green cross-sheet links); never hardcode inside a formula — every number lives in a labelled input; one consistent formula across each forecast row; separate inputs, calculations, and outputs; build error checks including a balance check and a master checks cell; consistent sign conventions and units; and clear documentation with an assumptions log and version history. The goal is a model anyone can open, understand, and trust.

Q46: Why is hardcoding inside formulas bad?

A: It hides assumptions where no one can find or audit them, so changing a rate means hunting through formulas instead of editing one input cell; it breaks the "one formula across the row" discipline; and it makes scenarios and sensitivities impossible because the driver isn't a referenceable cell. Put every changeable number in a blue input and reference it.

Q47: What colour conventions do you use and why?

A: Blue font for hardcoded inputs, black for formulas within the sheet, green for links to other sheets in the same workbook, and red for external-workbook links (which are fragile and worth flagging). It lets any reviewer instantly see what they can safely change (blue) versus what is calculated (black), which is essential for collaboration and error-spotting.

Q48: How do you make a large model calculate faster?

A: Minimise volatile functions (OFFSET, INDIRECT, NOW, TODAY, RAND); prefer INDEX/MATCH over full-column array formulas; avoid excessive data tables or set calculation to "Automatic Except Data Tables"; replace huge nested IFs with lookups; break unnecessary cross-workbook links; and switch to manual calculation (F9 to recalc) while building. Keep ranges tight rather than referencing entire columns.

Q49: A managing director says the model output "looks wrong". How do you respond?

A: Stay calm and investigate rather than defend. Check the assumptions driving the output first, run the balance and error checks, sanity-check margins/growth/terminal-value share against comparables and history, and trace precedents on the suspect number. Reproduce their expectation to see where the divergence starts. Then explain the driver behind the result — often the model is right but an assumption is aggressive, or there's a genuine link error to fix. Show the work transparently.

Q50: What units and conventions would you use for an Indian company model?

A: Typically ₹ crore for a large-cap (₹1 crore = ₹10 million), stated explicitly at the top, with per-share figures in rupees. Financials follow Ind AS; note items like reserves & surplus for retained earnings, and be careful with GST, and with consolidated vs standalone statements. Fiscal year usually ends 31 March, so label columns FY26E, FY27E, etc., and align with the company's reporting calendar.

Q51: What is the difference between EBIT, EBITDA, and free cash flow?

A: EBIT is operating profit after depreciation and amortisation. EBITDA adds D&A back to EBIT — a rough proxy for operating cash generation before capital-structure and non-cash items. Free cash flow goes further: $FCFF = EBIT \times (1 - \text{tax}) + D\&A - \text{capex} - \Delta NWC$, i.e. the cash available to all capital providers after reinvestment. EBITDA ignores capex and working capital, which is why it can overstate real cash flow for capital-intensive businesses.

Q52: How do you sanity-check a completed model?

A: Compare forecast margins, growth rates, returns (ROE/ROIC), and leverage against history and peers — a forecast that drifts far from the company's track record without a reason is suspect. Check that terminal value isn't an implausible share of DCF value, that revenue growth fades toward GDP over time, that the balance sheet balances in every year, and that the model behaves sensibly under stress inputs. Numbers should tell a coherent business story.

Q53: What's the difference between a budget/FP&A model and a valuation model?

A: An FP&A budget model is internal and operational — a detailed, driver-based annual plan and rolling forecast built for budget-vs-actual variance tracking and management decisions, usually monthly and bottom-up. A valuation model (DCF/LBO/M&A) is decision-specific and often annual, aimed at an external transaction or rating. The FP&A model prizes granularity and actuals integration; the valuation model prizes clean assumptions and clear outputs.

Q54: What does "one formula across the row" mean and why does it matter?

A: Every cell in a forecast row should contain the identical formula, just copied across periods, so the logic is consistent and auditable. If one column's formula differs, it's either a bug or an undocumented exception — both dangerous. It also makes the model easy to extend (drag right) and lets a reviewer check one cell and trust the whole row.

Q55: How do you handle taxes in a model, including deferred tax and NOLs?

A: At a basic level, $\text{tax} = \text{pre-tax income} \times \text{effective tax rate}$. More precisely, book tax differs from cash tax due to timing differences (e.g. accelerated depreciation for tax), captured in deferred tax assets/liabilities on the balance sheet. Net operating losses (carried-forward losses) shield future taxable income until exhausted, so you track an NOL balance and apply it before charging cash tax. In India, note MAT (minimum alternate tax) and the ability to carry forward business losses for eight years.

Q56: If net income is positive but the company is running out of cash, what could explain it?

A: Accrual profit isn't cash. Possible causes: rapid growth tying up cash in receivables and inventory (rising working capital), heavy capex exceeding operating cash flow, large debt repayments, aggressive revenue recognition booking sales before cash collection, or one-off non-cash gains inflating net income. The cash flow statement will reveal which — this is exactly why cash flow matters more than reported profit.

Q57: Why might interest be modelled on the beginning debt balance rather than the average?

A: To avoid circularity. Interest on the average balance depends on the ending balance, which depends on the cash generated, which depends on interest — a loop. Interest on the beginning balance depends only on last period's known number, breaking the loop, at the cost of slightly understating (or overstating) interest when debt changes materially within the year. It's a common, defensible simplification, especially in interviews.

Q58: What error checks would you hard-wire into every model?

A: A balance check ($\text{Assets} - \text{Liabilities} - \text{Equity} = 0$ each year); a cash tie ($\text{balance-sheet cash} = \text{cash-flow ending cash}$); schedule ties (PP&E, debt, equity roll-forwards equal their statement lines); sources = uses in a transaction; a "sum of segments = total" check for driver builds; and a single master cell that is TRUE only if all sub-checks pass, formatted red/green with conditional formatting so a problem is impossible to miss.

16.8 LBO, M&A & advanced Excel

Q59: Walk through the debt tranches in an LBO.

A: From senior to junior: the revolver (cheapest, secured, flexible, used for working-capital swings and as the plug); Term Loan A (bank-held, floating, amortising straight-line); Term Loan B (institutional, floating, longer tenor, minimal ~1%/yr mandatory amortisation with the rest swept); high-yield bonds (fixed coupon, higher cost, bullet at maturity, often non-call for a period); and mezzanine/subordinated (most expensive, sometimes PIK plus equity warrants, repaid last). Senior secured debt is cheapest and repaid first; junior debt costs more and is repaid last, reflecting its risk.

Q60: What is a cash sweep?

A: A mechanism that directs the company's surplus free cash flow — after interest and mandatory amortisation — to prepay debt early, in seniority order (revolver, then TLA, TLB, bonds). A 100% sweep applies all excess cash. It accelerates deleveraging: less debt means less interest next year, freeing even more cash to sweep — the deleveraging flywheel that drives LBO equity returns. It also interacts with interest, producing circularity like the three-statement model.

Q61: What is PIK interest and why use it?

A: Payment-in-kind interest accrues onto the loan principal instead of being paid in cash. It preserves near-term cash flow (helpful for a highly levered company or to fund growth) at the cost of a compounding, larger balance repaid later. Common on mezzanine and holdco notes. In the model, PIK increases the debt balance each period rather than hitting the cash flow.

Q62: Explain a returns/value-creation bridge in an LBO.

A: It attributes the equity gain to three sources: EBITDA growth ($\Delta \text{EBITDA} \times \text{entry multiple}$), multiple expansion ($\Delta \text{Multiple} \times \text{exit EBITDA}$), and debt paydown/free-cash-flow generation ($\Delta \text{net debt}$). They sum to the total equity value gain. A high-quality deal earns most of its return from operational EBITDA growth and deleveraging rather than betting on a higher exit multiple, which the sponsor cannot control.

Q63: Buy at 8× EBITDA of ₹500 cr with ₹2,400 cr debt; in 5 years EBITDA is ₹700 cr, net debt ₹1,000 cr, exit at 8×. IRR and MOIC?

A: Entry EV = $8 \times 500 = ₹4,000$ cr; entry equity = $4,000 - 2,400 = ₹1,600$ cr. Exit EV = $8 \times 700 = ₹5,600$ cr; exit equity = $5,600 - 1,000 = ₹4,600$ cr. MOIC = $4,600 / 1,600 = 2.875\times$. IRR = $2.875^{(1/5)} - 1 \approx 23.5\%$. Bridge: EBITDA growth (200×8) = ₹1,600; multiple o; debt paydown ($2,400 - 1,000$) = ₹1,400; total ₹3,000 cr gain.

Q64: How is goodwill created in an acquisition?

A: Under acquisition accounting (Ind AS 103/IFRS 3), the acquirer records the target's assets and liabilities at fair value, stepping up PP&E and recognising new intangibles (brands, customer lists). Goodwill = equity purchase price – fair value of net identifiable assets acquired. It is the residual that makes the pro-forma balance sheet balance. Under Ind AS/IFRS goodwill isn't amortised but is tested annually for impairment; finite-life intangibles are amortised, adding expense that dents pro-forma EPS.

Q65: A target has ₹1,500 cr book equity; you pay ₹4,000 cr; step-ups add ₹1,000 cr of assets and create a ₹250 cr DTL. Goodwill?

A: Fair value of net identifiable assets = $1,500 + 1,000 - 250 = ₹2,250$ cr. Goodwill = $4,000 - 2,250 = ₹1,750$ cr. The deferred tax liability arises because the step-up is recognised for book but not for tax, so future book depreciation/amortisation on the step-up isn't tax-deductible.

Q66: What are synergies, and how do cost and revenue synergies differ?

A: Synergies are the incremental value from combining two firms, justifying the control premium. Cost synergies (removing duplicate headcount, procurement scale, closing facilities) are credible and readily modelled. Revenue synergies (cross-selling, wider distribution) are real but softer, so analysts haircut them heavily. Model both net of the one-time cost-to-achieve and phase them in over 2–3 years; add after-tax synergies to pro-forma net income in an accretion/dilution analysis.

Q67: A ₹3,000 cr all-debt deal at 8% (25% tax) adds ₹150 cr target NI; acquirer EPS is ₹10 on 100 cr shares.**Accretive? Breakeven synergies?**

A: After-tax interest = $3,000 \times 8\% \times 0.75 = ₹180$ cr. Pro-forma NI = $1,000 + 150 - 180 = ₹970$ cr on 100 cr shares → EPS ₹9.70, dilutive by ₹0.30. Required after-tax synergy = $(10 \times 100) - 970 = ₹30$ cr; pre-tax = $30/0.75 = ₹40$ cr. Above ₹40 cr pre-tax synergies the deal turns accretive. You can also solve this with Goal Seek on the synergy cell.

Q68: When is a cash- or debt-funded deal accretive?

A: When the target's after-tax earnings yield exceeds the after-tax cost of the funding. For cash, the "cost" is the foregone after-tax interest income on that cash; for debt, it's the after-tax interest rate. If the incremental earnings you buy exceed the incremental cost you incur, EPS rises. Because cash and debt are usually "cheaper" than equity, cash/debt deals are more often accretive than all-stock deals — but they add leverage/reduce liquidity.

Q69: How does Goal Seek differ from Solver?

A: Goal Seek changes one input to make one output hit a target, with no constraints (e.g. find the WACC that makes DCF value = market price, or the entry multiple that gives a 25% IRR). Solver is an add-in that optimises an objective — maximise, minimise, or target — by changing multiple inputs subject to constraints (e.g. maximise IRR subject to a leverage cap and a minimum DSCR). Use Goal Seek for simple back-solves, Solver for constrained multi-variable optimisation.

Q70: Set up a two-variable data table for a DCF share price vs WACC and terminal growth.

A: In the top-left corner cell of a blank grid, put a link to the output: =SharePriceCell. List WACC values down the left column below the corner, and g values across the top row to the right. Select the whole rectangle, then Data → What-If Analysis → Data Table; set Row input cell = the g input (it runs across the row) and Column input cell = the WACC input (it runs down the column). Excel fills the grid with {=TABLE(g, WACC)}. Getting the row/column inputs backwards is the classic mistake.

Q71: What are dynamic array functions, and when would you use FILTER or UNIQUE?

A: Excel 365 functions that "spill" a whole result array from one formula. UNIQUE(range) returns the distinct values — handy to build a driver or segment list from a raw data dump. FILTER(array, condition) returns only the rows meeting a condition, e.g. all transactions for one segment. SORT and SEQUENCE round out the set. You reference the spilled range with the # operator, e.g. =SUM(F2#). They replace many array-formula and helper-column hacks.

Q72: What do LET and LAMBDA do?

A: LET lets you name intermediate results inside a single formula — =LET(rev, A1*A2, marg, 0.2, rev*marg) — improving readability and speed since each named part computes once. LAMBDA lets you define a reusable custom function without VBA; saved via Name Manager it becomes callable like a built-in. Together they bring lightweight programming into the grid; know them conceptually as the modern alternative to sprawling nested formulas and VBA UDFs.

Q73: What is Power Query and why use it?

A: Power Query (Data → Get & Transform) is Excel's ETL tool. It connects to files, folders, databases or the web and records a repeatable sequence of cleaning steps — remove columns, unpivot, merge, change types, filter. When the source updates you just click Refresh and the cleaned table rebuilds, so you never re-do a manual clean. It's the professional way to feed a model from messy trial-balance or blotter exports and keeps raw data out of the calculation sheets.

Q74: How would you build a loan amortisation schedule in Excel?

A: Compute the level payment with $\text{PMT}(\text{rate}, \text{nper}, \text{pv})$, then for each period split it with $\text{IPMT}(\text{rate}, \text{per}, \text{nper}, \text{pv})$ (interest portion) and $\text{PPMT}(\text{rate}, \text{per}, \text{nper}, \text{pv})$ (principal portion) — $\text{IPMT} + \text{PPMT} = \text{PMT}$. Roll the balance: $\text{ending} = \text{beginning} - \text{principal paid}$. Interest falls and principal rises over the life. This is exactly how you'd model an amortising term loan in a debt schedule, feeding interest to the P&L and principal to financing cash flow.

Q75: What's the difference between IRR and MIRR, and when does IRR mislead?

A: IRR assumes interim cash flows are reinvested at the IRR itself, which is often unrealistic, and it can return multiple values when cash flows change sign more than once. MIRR fixes both: $\text{MIRR}(\text{values}, \text{finance_rate}, \text{reinvest_rate})$ reinvests positives at a stated reinvestment rate and discounts negatives at a financing rate, giving a single, more realistic figure. For dated, irregular flows, XIRR is the right tool; for conventional single-sign-change flows, plain IRR is fine.

Q76: Convert a 12% nominal rate compounded monthly to an effective annual rate.

A: $\text{=EFFECT}(12\%, 12) \rightarrow 12.68\%$. Formula: $\text{EAR} = (1 + 0.12/12)^{12} - 1 = 12.68\%$. EFFECT/NOMINAL let you compare loans quoted on different compounding bases apples-to-apples — essential before deciding which financing is genuinely cheaper.

Key takeaways for the interview: Be able to (1) walk through a three-statement model end-to-end, (2) trace any single transaction through all three statements, (3) explain balancing, plugs, and debugging, (4) discuss circularity and the switch, (5) quote correct syntax for INDEX/MATCH, SUMIFS, NPV/XNPV, IRR/XIRR, and data tables, and (6) give the intuition behind DCF, LBO, and accretion/dilution. Precision and calm structure beat speed.

17. Final Checklist & Key Takeaways

17.1 The 60-second model walkthrough (memorise)

1. Historicals in and tied out → schedules built off assumptions.
2. P&L to EBIT; balance-sheet items from schedules; cash flow derived from NI + non-cash – Δ balance-sheet accounts.
3. Ending cash → balance-sheet cash; interest → P&L (circularity); revolver as the plug.
4. Balance check = 0 every year; master checks cell TRUE.

17.2 Non-negotiable disciplines

- Blue inputs, black formulas, green links. No hardcodes in formulas. One formula per row.
- Cash comes from the cash flow statement — never a hardcode.
- Watch for 2×-a-line-item imbalance (sign error) and 1×-a-line-item imbalance (missing link).
- Know your interest convention (average = circular, beginning = clean) and keep a circularity switch.
- NPV excludes time-0 flow; use XNPV/XIRR for dated cash flows.
- Data table: output link in the corner; row-input for the top-row variable, column-input for the left-column variable.

17.3 The one-line summaries per model

- **Three-statement:** integrated forecast; the foundation of everything.
- **DCF:** value = PV of future free cash flows + PV of terminal value.
- **LBO:** buy with debt, deleverage, exit; returns from paydown + growth + multiple.
- **M&A:** accretive if you pay a lower P/E than your own (all-stock).
- **Budget/FP&A:** operational plan for variance tracking.

Final word: The interviewer is testing whether you understand the plumbing and whether your Excel is disciplined. Explain links precisely, trace transactions calmly, quote function syntax correctly, and always come back to the balance check. Do that and you will present as someone who has genuinely built models — which, after two years on a derivatives desk and with NISM Series-XV behind you, you can absolutely be.